

PUBLIC SMART CONTRACT AUDIT REPORT

# for Exactly Protocol

Oracle Staleness -- Lending Protocol

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|----------------|--------------------|
| Prepared By    | Red Ghost Security |
| Date           | May 9, 2026        |
| Classification | Public             |
| Report #       | RGS-2026-001       |

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|----------------|--------------------------------------|
| Client         | Exactly Protocol                     |
| Title          | Smart Contract Audit Report          |
| Target         | Oracle Staleness -- Lending Protocol |
| Version        | 1.0                                  |
| Chain          | Optimism Mainnet                     |
| Author         | Red Ghost 47                         |
| Auditor        | Red Ghost 47                         |
| Classification | Public                               |
| Report #       | RGS-2026-001                         |

| Version | Date        | Author(s)    | Description   |
|---------|-------------|--------------|---------------|
| 1.0     | May 9, 2026 | Red Ghost 47 | Final Release |

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## 1. Introduction

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## 1.1 About the Audited Protocol

Red Ghost Security was engaged to perform a security audit of the Exactly Protocol protocol, a Lending Protocol on Optimism Mainnet. The audit was conducted between May 8, 2026 and May 9, 2026.

The research methodology included manual code review, Foundry-based invariant fuzz testing, and exploit scenario simulation on a mainnet fork.

## 1.2 About Red Ghost Security

Red Ghost Security is a DeFi smart contract audit and exploit research firm. We specialize in oracle manipulation analysis, flash loan attack surface review, and zero-day discovery. Every finding is validated as a working PoC on a mainnet fork with precise economic impact measurement.

## 1.3 Audit Methodology

### Manual Code Review

Line-by-line Solidity analysis for logical flaws and security vulnerabilities.

### Static Analysis

Automated scanning using Slither and custom detectors.

### Foundry Fuzz Testing

Invariant-based fuzz testing to identify edge cases.

### Exploit Simulation

Each finding becomes a working PoC on a mainnet fork.

## 1.4 Disclaimer

*This report does not constitute a guarantee or warranty that the audited contracts are free from all vulnerabilities. The audit was performed on a specific version of the codebase. Red Ghost Security shall not be held liable for losses incurred from using these contracts.*

### 2.1 Executive Summary

A total of 1 finding were identified. These include 1 critical.

### 2.2 Severity Breakdown

|          |   |
|----------|---|
| Critical | 1 |
|----------|---|

| ID      | Title                                                | Severity | Status |
|---------|------------------------------------------------------|----------|--------|
| RGS-005 | Oracle Staleness -- latestAnswer() Without Freshness | Critical | Open   |

**RGS-005 -- CRITICAL****Description**

The protocol uses Chainlink's `latestAnswer()` without checking the timestamp of the last update. If the oracle becomes stale (network congestion, Chainlink node issues, L1 sequencer downtime on Optimism), the protocol continues using outdated prices.

**Impact**

Stale prices can persist for hours. Users can execute liquidations or borrowings based on prices that no longer reflect market conditions, leading to undercollateralized loans or unfair liquidations.

**Exploit Scenario**

1. Chainlink oracle update delayed on Optimism. 2. Price used is from 4+ hours ago. 3. Attacker borrows against assets that dropped significantly in value. 4. Protocol only discovers correct price when oracle updates. 5. By then, attacker has extracted value.

**Recommendation**

Replace `latestAnswer()` with `latestRoundData()` and validate the `updatedAt` timestamp. Set maximum staleness threshold (e.g., 1 hour). Check L1 sequencer status using Chainlink's L1 Sequencer Uptime Feed on Optimism.

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Our security analysis of the Exactly Protocol protocol identified 1 finding, including 1 critical-severity and 0 high-severity issues. Remediation recommendations have been communicated to the team.

Red Ghost Security recommends addressing all findings before mainnet deployment. A follow-up audit is recommended to verify remediation.

Red Ghost Security

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